

A counterexample and a sharp spacing-three criterion for products of q -integers

Prepared with ChatGPT; self-contained proofs and exact computational checks

Abstract

We attack Conjecture 5.4 of Connelly, Ito, Martinez, Shevchenko, and Yang, *Unimodality of q -Fibonomial coefficients for small cases*, arXiv:2605.12822v1. Its necessity assertion is false: $(1+q)^6(1+q^3)$ is unimodal but violates the proposed condition. We then prove the conjectured sufficient condition for every spacing $r \geq 2$. For spacing three we obtain a complete replacement: if no a_i is divisible by three, then

$$\left(\prod_i [a_i]_q \right) [b]_{q^3} \text{ is unimodal} \iff b \leq 1 + \sum_i \left\lfloor \frac{a_i}{3} \right\rfloor + 2 \left\lfloor \frac{\#\{i : a_i \equiv 2 \pmod{3}\}}{6} \right\rfloor.$$

A failed inequality produces a specified first-half coefficient drop of exactly one. We also establish a general necessary degree bound, settle the spacing-two criterion, and prove the sharp two-shift threshold $(1+q)^n(1+q^r)$ unimodal if and only if $n \geq r^2 - 3$. The arguments are finite and algebraic; computations are supplementary. The main q -Fibonomial unimodality conjecture is not resolved here.

The counterexample at a glance

$$(1+q)^6(1+q^3) = 1 + 6q + 15q^2 + 21q^3 + 21q^4 + 21q^5 + 21q^6 + 15q^7 + 6q^8 + q^9.$$

Here $r = 3$, $b = 2$, and $a_1 = \dots = a_6 = 2$. No a_i is divisible by 3, and $1 + \sum_i \lfloor a_i/3 \rfloor = 1 < 2$. Nevertheless, the displayed coefficient sequence is weakly unimodal.

Status. The statements below are accompanied by proofs and reproducible integer-arithmetic tests. They have not been independently refereed or checked by a proof assistant. The counterexample refutes the specified source statement; historical priority of the counterexample and the stronger theorems has not been established by an exhaustive literature review.

1 The selected conjecture and the result

For a positive integer a , put

$$[a]_q = 1 + q + \dots + q^{a-1}.$$

Throughout, q is a formal indeterminate, $r \geq 2$, $k \geq 1$, and $a_1, \dots, a_k, b$ are positive integers. We study

$$P_{a,b,r}(q) = \left(\prod_{i=1}^k [a_i]_q \right) [b]_{q^r}. \quad (1.1)$$

Its coefficients are nonnegative integers, its constant and leading coefficients are one, and its degree is

$$D = S + r(b-1), \quad S = \sum_{i=1}^k (a_i - 1). \quad (1.2)$$

Every factor is symmetric, so P is symmetric.

The source of the attack is [1, Conjecture 5.4, p. 14]. For brevity write its proposed condition as

$$C_r(a, b) : (\exists i: r \mid a_i) \quad \text{or} \quad b \leq 1 + \sum_i \left\lfloor \frac{a_i}{r} \right\rfloor. \quad (1.3)$$

The conjecture asserts that (1.3) is sufficient, and also necessary when $k \leq 3$ or $r \leq 3$. The distinction between “or” and “and” matters: the case $k = 6, r = 3$ lies inside the asserted necessity range. The source reports tests with $k \leq 5, r \leq 6$, and $\max\{a_i, b\} \leq 15$ [1, p. 14].

Proposition 1.1 (Exact disproof). *The necessity assertion in [1, Conjecture 5.4] is false.*

Proof. Take $r = 3, b = 2$, and $a_i = 2$ for $1 \leq i \leq 6$. The binomial theorem gives the coefficient list

$$(1, 6, 15, 21, 21, 21, 15, 6, 1).$$

It increases weakly to its central plateau and then decreases weakly. On the other hand, $3 \nmid 2$ and $b = 2 > 1 + 6\lfloor 2/3 \rfloor = 1$, so (1.3) fails. Since $r = 3$, the claimed necessity applies to this example. $\square$

This is a counterexample to an auxiliary product criterion, not to the main q -Fibonomial conjecture in [1]. Our stronger conclusions, proved below, are summarized in Table 1. All appearances of “unimodal” allow equal consecutive coefficients.

Setting	Conclusion proved in this note
Every $r \geq 2$	The condition C_r is sufficient. With no divisible factor, unimodality also requires $b \leq 1 + \lfloor S/r \rfloor$.
$r = 2$	The original condition C_2 is necessary and sufficient.
$r = 3$	With no divisible factor, the exact upper bound on b is $1 + \sum_i \lfloor a_i/3 \rfloor + 2\lfloor t/6 \rfloor$, where $t = \#\{i : a_i \equiv 2 \pmod{3}\}$.
Two shifted binomial rows	$(1+q)^n(1+q^r)$ is unimodal exactly when $n \geq r^2 - 3$, for $r \geq 2, n \geq 0$.

Table 1: Mathematical results, rather than extrapolations from the finite tests.

2 Elementary tools for symmetric unimodality

A finite nonnegative sequence $(c_0, \dots, c_D)$ is *unimodal* if, for some index m , it is weakly increasing up to m and weakly decreasing after m . It is *symmetric* if $c_j = c_{D-j}$. For a symmetric sequence, unimodality is equivalent to

$$c_j - c_{j-1} \geq 0 \quad 0 \leq j \leq \lfloor D/2 \rfloor, \quad c_{-1} = 0. \quad (2.1)$$

We abbreviate “symmetric and unimodal” by SU. All coefficient sequences are extended by zero outside their support.

A shifted sequence $q^s H(q)$ has its center shifted by s . In a sum of shifted polynomials, symmetry is understood about the *specified common center*, not automatically half the largest exponent of each summand. This convention is essential in the decomposition used below.

Lemma 2.1 (Centered intervals and closure). *Products of SU polynomials with nonnegative coefficients are SU. Nonnegative sums of SU coefficient sequences with the same center are also SU.*

Proof. If $F(q) = \sum_{j=0}^D f_j q^j$ is SU, let $\alpha_j = f_j - f_{j-1}$ for $0 \leq j \leq \lfloor D/2 \rfloor$, with $f_{-1} = 0$. Then $\alpha_j \geq 0$ and

$$F(q) = \sum_{j=0}^{\lfloor D/2 \rfloor} \alpha_j q^j [D - 2j + 1]_q. \quad (2.2)$$

To verify the identity, the coefficient at q^m on the right is $\sum_{j=0}^{\min(m, D-m)} \alpha_j = f_{\min(m, D-m)} = f_m$. Each summand is the indicator sequence of an interval centered at $D/2$.

For positive u, v , the coefficients of $[u]_q[v]_q$ at exponents $0 \leq \ell \leq u + v - 2$ are

$$\min\{\ell + 1, u, v, u + v - 1 - \ell\}.$$

They form a symmetric triangular or trapezoidal sequence. Apply (2.2) to two factors of degrees D and E . Every product of two interval summands is SU with center $(D + E)/2$. On the left of this center, each first difference is nonnegative, so the same is true of their sum. Symmetry supplies the right half. This also proves the assertion about sums with a common center. $\square$

Lemma 2.2 (A divisible factor). *If $r \mid a_i$ for some i , then $P_{a,b,r}$ is unimodal for every positive integer b .*

Proof. Write $a_i = rA$. The identity

$$[rA]_q = [r]_q[A]_{q^r}$$

shows that the relevant two factors are

$$[r]_q[A]_{q^r}[b]_{q^r}.$$

The polynomial $[A]_z[b]_z$ has an SU coefficient sequence in z . Replacing z by q^r and multiplying by $[r]_q$ repeats each coefficient exactly r times. Repetition preserves symmetry and unimodality. The remaining factors are ordinary q -integers, so Lemma 2.1 finishes the proof. $\square$

3 A centered identity proves the sufficient condition

The following identity removes one block of length r from an ordinary factor and one term from the spaced factor, without changing the center of the residual sequence.

Lemma 3.1 (Centered peeling). *For integers $a > r$ and $b \geq 2$,*

$$[a]_q[b]_{q^r} = [a + r(b - 1)]_q + q^r[a - r]_q[b - 1]_{q^r}. \quad (3.1)$$

Both terms on the right, with zero padding, are symmetric about the center of the polynomial on the left.

Proof. Multiply (3.1) by $(1 - q)(1 - q^r)$. The identity becomes

$$\begin{aligned} (1 - q^a)(1 - q^{rb}) &= (1 - q^{a+r(b-1)})(1 - q^r) \\ &\quad + q^r(1 - q^{a-r})(1 - q^{r(b-1)}), \end{aligned}$$

which follows on expansion. The left side of (3.1) has polynomial degree

$$d = a - 1 + r(b - 1).$$

The first term on the right has degree d . The unshifted residual polynomial has degree $d - 2r$, so its shift by q^r has center $r + (d - 2r)/2 = d/2$. $\square$

Theorem 3.2 (Sufficiency for every spacing). *For every $r \geq 2$ and every number k of ordinary factors, $C_r(a, b)$ implies unimodality of $P_{a,b,r}$.*

Proof. A divisible factor is covered by Lemma 2.2. Assume there is none, and write

$$Q = \sum_i \left\lfloor \frac{a_i}{r} \right\rfloor.$$

We prove sufficiency of $b \leq Q + 1$ by induction on b . For $b = 1$, the spaced factor equals one and the product of ordinary q -integers is SU by Lemma 2.1.

Let $b \geq 2$. Then $Q \geq 1$, so at least one a_i is greater than r : equality is impossible because there is no divisible factor. Apply (3.1) to this factor and multiply by all the others. The first summand is a product of ordinary q -integers, hence SU. In the residual product, a_i is replaced by $a_i - r$ and b by $b - 1$. No new divisible factor appears, the floor sum becomes $Q - 1$, and

$$b - 1 \leq Q = (Q - 1) + 1.$$

The induction hypothesis makes the residual product SU. Lemma 3.1 shows that the shifted residual and the first summand have the same center, even after multiplication by the other factors. Their sum is therefore SU by Lemma 2.1. $\square$

Corollary 3.3 (Reduction to small residues). *Suppose no a_i is divisible by r , and write*

$$a_i = rA_i + s_i, \quad 1 \leq s_i \leq r - 1, \quad Q = \sum_i A_i.$$

If $b > Q + 1$ and

$$\left(\prod_i [s_i]_q \right) [b - Q]_{q^r} \tag{3.2}$$

is SU, then $P_{a,b,r}$ is SU.

Proof. Apply (3.1) a total of Q times, subtracting r from some a_i at each step. The spaced parameter never drops below two, since its final value is $b - Q \geq 2$. The final residual is (3.2). Reversing the steps, each step adds an SU product of ordinary q -integers at the same center as the shifted residual. $\square$

Corollary 3.3 is a sufficient reduction, not an assertion that residue reduction preserves unimodality in both directions for every r . That converse will be obtained for $r = 3$ by a separate obstruction argument.

4 Periodic coefficient differences and a necessary bound

Let

$$F(q) = \prod_i [a_i]_q = \sum_{j=0}^S f_j q^j, \quad R_s = \sum_{\substack{0 \leq j \leq S \\ j \equiv s \pmod{r}}} f_j, \quad 0 \leq s < r.$$

The R_s are the total coefficient masses in the residue classes modulo r . Put $\delta_s = R_s - R_{s-1}$, with subscripts interpreted modulo r . As a formal power series, define

$$U(q) = \frac{F(q)}{1 - q^r} = \sum_{n \geq 0} u_n q^n, \quad u_n = \sum_{\ell \geq 0} f_{n - r\ell}. \tag{4.1}$$

As usual, coefficients outside their support are zero.

Lemma 4.1 (The periodic tail). *If $n \geq \max\{1, S - r + 2\}$, then $u_n - u_{n-1} = \delta_{n \bmod r}$. If also $n < rb$, the same difference occurs in $P = F[b]_{q^r}$:*

$$[q^n]P - [q^{n-1}]P = \delta_{n \bmod r}. \tag{4.2}$$

Proof. For the stated n , the next possible omitted indices in the residue classes of n and $n - 1$ are $n + r > S$ and $n - 1 + r > S$. Consequently $u_n = R_{n \bmod r}$ and $u_{n-1} = R_{(n-1) \bmod r}$. Finally,

$$P(q) = (1 - q^{rb})U(q),$$

so its coefficients agree with those of U below exponent rb . $\square$

Theorem 4.2 (A necessary degree bound for every spacing). *Suppose no a_i is divisible by r . If $P_{a,b,r}$ is unimodal, then*

$$r(b - 1) \leq S, \quad \text{equivalently} \quad b \leq 1 + \left\lfloor \frac{S}{r} \right\rfloor. \quad (4.3)$$

Proof. Let ζ be a primitive r th root of unity. Every factor

$$[a_i]_q \Big|_{q=\zeta} = \frac{1 - \zeta^{a_i}}{1 - \zeta}$$

is nonzero, so $F(\zeta) \neq 0$. The R_s cannot all be equal, since otherwise $F(\zeta) = R_0(1 + \zeta + \dots + \zeta^{r-1}) = 0$. The differences δ_s sum to zero and are not all zero, so at least one is negative.

Choose an integer n in the r -term interval

$$S - r + 2 \leq n \leq S + 1$$

whose residue has a negative difference. Such an n is automatically positive. Indeed, if the interval includes nonpositive integers then $S \leq r - 2$; at $n = 0$ the difference is $R_0 - R_{r-1} = f_0 > 0$, and for any negative n in this interval both relevant residue masses vanish. Thus Lemma 4.1 applies.

Suppose for a contradiction that $L = r(b - 1) > S$. Then $n \leq S + 1 < rb$. If $L \geq S + 2$, we also have $n \leq S + 1 \leq \lfloor (S + L)/2 \rfloor$. Equation (4.2) gives a negative first-half difference, contradicting unimodality.

The remaining possibility is $L = S + 1$. In that case $S \equiv -1 \pmod{r}$. Symmetry of F implies $R_0 = R_{r-1}$, hence $\delta_0 = 0$. The selected negative difference cannot be at $n = S + 1$, whose residue is zero. Therefore $n \leq S = \lfloor (S + L)/2 \rfloor$, and again (4.2) contradicts unimodality. $\square$

Together, Theorems 3.2 and 4.2 give a useful sandwich. When there is no divisible factor,

$$b \leq 1 + \sum_i \left\lfloor \frac{a_i}{r} \right\rfloor \implies P \text{ unimodal} \implies b \leq 1 + \left\lfloor \frac{\sum_i (a_i - 1)}{r} \right\rfloor. \quad (4.4)$$

The residue classes decide what happens in the possible gap.

5 Spacing two: the original criterion is exact

Theorem 5.1 (Complete spacing-two criterion). *The polynomial $(\prod_i [a_i]_q)[b]_{q^2}$ is unimodal if and only if some a_i is even or*

$$b \leq 1 + \sum_i \left\lfloor \frac{a_i}{2} \right\rfloor.$$

Proof. Sufficiency follows from Theorem 3.2. If all a_i are odd, write $a_i = 2A_i + 1$ and $Q = \sum_i A_i$. Then $S = 2Q$, so Theorem 4.2 gives the same necessary bound $b \leq Q + 1$. $\square$

There is also an exact failure certificate. With all a_i odd, $F(-1) = 1$, so the even residue mass exceeds the odd residue mass by one. For $b \geq Q + 2$, the index $n = S + 1 = 2Q + 1$ lies below $2b$ and no later than the center $Q + b - 1$. Lemma 4.1 gives

$$[q^{2Q+1}]P - [q^{2Q}]P = -1.$$

This argument also covers factors $a_i = 1$.

6 Spacing three: a complete replacement theorem

Assume initially that no a_i is divisible by three. Define

$$a_i = 3A_i + s_i, \quad s_i \in \{1, 2\}, \quad Q = \sum_i A_i, \quad t = \#\{i : s_i = 2\}. \quad (6.1)$$

Then

$$S = 3Q + t. \quad (6.2)$$

Theorem 6.1 (Sharp spacing-three criterion). *The polynomial $P_{a,b,3}$ is unimodal if and only if*

$$(\exists i : 3 \mid a_i) \quad \text{or} \quad b \leq 1 + Q + 2 \left\lfloor \frac{t}{6} \right\rfloor, \quad (6.3)$$

where Q, t are as in (6.1) in the absence of a divisible factor. If no a_i is divisible by three and the inequality fails, write $t = 6h + v$, $0 \leq v \leq 5$, and $M = Q + 2h$. At the explicit index

$$n_0 = 3M + \left\lfloor \frac{v+3}{2} \right\rfloor \quad (6.4)$$

we have

$$n_0 \leq \lfloor D/2 \rfloor, \quad [q^{n_0}]P - [q^{n_0-1}]P = -1. \quad (6.5)$$

The proof has two independent parts. A binomial-prefix lemma proves sufficiency. A six-case residue table proves necessity, including the unit-drop certificate.

6.1 The binomial-prefix lemma

Lemma 6.2 (Enough ordinary binary factors). *For integers $B \geq 1$ and $t \geq 0$, the polynomial*

$$(1+q)^t [B]_{q^3}$$

is unimodal whenever $t \geq 6\lfloor B/2 \rfloor$.

Proof. For $B = 1$ this is an ordinary binomial polynomial. Suppose $B \geq 2$, put $m = \lfloor B/2 \rfloor$ and $K = 6m$, and first take $t = K$. Thus $B = 2m$ or $B = 2m + 1$. Let

$$f_j = \binom{K}{j}, \quad u_n = \sum_{\ell \geq 0} f_{n-3\ell}, \quad d_n = u_n - u_{n-1},$$

with $f_j = 0$ outside $0 \leq j \leq K$ and $u_n = d_n = 0$ for $n < 0$. Let R_0, R_1, R_2 be the full residue masses of $(1+q)^K$.

Modulo $\Phi_3(q) = 1 + q + q^2$, one has $(1+q)^3 \equiv -1$ and hence $(1+q)^6 \equiv 1$. Since $6 \mid K$,

$$R_0 + R_1q + R_2q^2 \equiv 1 \pmod{\Phi_3(q)}.$$

Two polynomials of degree at most two congruent modulo Φ_3 differ by a constant multiple of Φ_3 . It follows that $R_0 - R_1 = 1$ and $R_1 = R_2$. Since $R_0 + R_1 + R_2 = 2^K$,

$$R_0 = \frac{2^K + 2}{3}, \quad R_1 = R_2 = \frac{2^K - 1}{3}. \quad (6.6)$$

Consequently

$$(\delta_0, \delta_1, \delta_2) = (1, -1, 0). \quad (6.7)$$

Symmetry $f_j = f_{K-j}$ gives a useful reflection identity. The terms of residue n omitted from u_n have indices $n+3, n+6, \dots$. Reflecting these indices about $K/2$ gives

$$u_n = R_{n \bmod 3} - u_{K-n-3} \quad (n \geq -1).$$

Subtract the same identity at $n-1$. For $n \geq 0$ this yields

$$d_n = \delta_{n \bmod 3} + d_{K-n-2}. \quad (6.8)$$

For $0 \leq n \leq K/2$,

$$d_n = \sum_{\ell \geq 0} (f_{n-3\ell} - f_{n-1-3\ell}) \geq 1.$$

Every summand is nonnegative because the binomial coefficients increase up to $K/2$: explicitly, $f_j/f_{j-1} = (K-j+1)/j > 1$ for $1 \leq j \leq K/2$. The summand with $\ell = 0$ is a positive integer. For $K/2 < n \leq K-2$, the reflected index $K-n-2$ lies between zero and $K/2$. Equations (6.7)–(6.8) therefore give $d_n \geq -1 + 1 = 0$. At the last two indices,

$$d_{K-1} = \delta_2 = 0, \quad d_K = \delta_0 = 1.$$

We have proved

$$d_n \geq 0 \quad (0 \leq n \leq K). \quad (6.9)$$

Now

$$(1+q)^K [B]_{q^3} = (1-q^{3B}) \sum_{n \geq 0} u_n q^n.$$

If $B = 2m$, its degree is $2K-3$, its left-middle index is $K-2$, and $3B = K$. If $B = 2m+1$, its degree is $2K$, its middle index is K , and $3B = K+3$. In either case the coefficients through the center agree with u_n , and their differences are nonnegative by (6.9). Symmetry proves unimodality for $t = K$. Finally, multiplication by $(1+q)^{t-K}$ preserves SU, proving the claim for every $t \geq K$. $\square$

6.2 Sufficiency of the corrected criterion

Proof of Theorem 6.1, sufficient direction. A divisible factor is handled by Lemma 2.2. Otherwise assume the right-hand inequality in (6.3). For $b \leq Q+1$, Theorem 3.2 applies. For $b > Q+1$, use Corollary 3.3 with $r = 3$. The residue product is

$$(1+q)^t [b-Q]_{q^3}.$$

Set $B = b-Q$. The assumed inequality says $B \leq 1+2\lfloor t/6 \rfloor$, which implies $t \geq 6\lfloor B/2 \rfloor$. Lemma 6.2 proves that the residue product is SU, and Corollary 3.3 lifts this back to the original product. $\square$

6.3 Necessity and the coefficient-drop certificate

Proof of Theorem 6.1, necessary direction. Suppose no a_i is divisible by three, and write $t = 6h + v$, $0 \leq v \leq 5$, and $M = Q + 2h$. Then $S = 3M + v$ by (6.2).

Let R_0, R_1, R_2 be the residue masses of $F = \prod_i [a_i]_q$ modulo three. Complete blocks of three terms vanish modulo $\Phi_3(q) = 1 + q + q^2$, so

$$F(q) \equiv \prod_i [s_i]_q = (1+q)^t \equiv (1+q)^v \pmod{\Phi_3(q)}. \quad (6.10)$$

The degree-at-most-two residue polynomials of F and $(1+q)^v$ are congruent modulo Φ_3 , so they differ by a constant multiple of Φ_3 . Their corresponding residue differences are therefore equal. The binomial theorem for $v = 0, \dots, 5$ gives Table 2.

v	δ_0	δ_1	δ_2	$e = \lfloor (v+3)/2 \rfloor$	$n_0 \bmod 3$	$n_0 - S$
0	1	-1	0	1	1	1
1	1	0	-1	2	2	1
2	0	1	-1	2	2	0
3	-1	1	0	3	0	0
4	-1	0	1	3	0	-1
5	0	-1	1	4	1	-1

Table 2: The residue class selected by $n_0 = 3M + e$ always has difference -1 .

If the bound in (6.3) fails, then $b \geq M + 2$. Choose n_0 from (6.4). The table shows $n_0 \geq S - 1$ and $n_0 \geq 1$. Moreover,

$$n_0 \leq 3M + 4 < 3M + 6 \leq 3b.$$

Thus Lemma 4.1 applies at n_0 , and the selected table entry gives the exact difference -1 . Finally,

$$\begin{aligned} \left\lfloor \frac{D}{2} \right\rfloor &= \left\lfloor \frac{3M + v + 3(b-1)}{2} \right\rfloor \\ &\geq \left\lfloor \frac{6M + v + 3}{2} \right\rfloor = 3M + \left\lfloor \frac{v+3}{2} \right\rfloor = n_0. \end{aligned}$$

The negative difference occurs in the first half of the symmetric sequence, contradicting (2.1). This proves both necessity and (6.5). $\square$

6.4 Consequences, minimality, and the missing sixth factor

Corollary 6.3 (A two-parameter infinite family). *For $t \geq 0$ and $B \geq 1$,*

$$(1+q)^t [B]_{q^3} \text{ is unimodal} \iff t \geq 6 \left\lfloor \frac{B}{2} \right\rfloor. \quad (6.11)$$

Proof. For $t \geq 1$, take t ordinary factors $a_i = 2$ in Theorem 6.1, so $Q = 0$. The condition becomes $B \leq 1 + 2\lfloor t/6 \rfloor$, equivalent to (6.11). For $t = 0$, the spaced polynomial is unimodal only for $B = 1$; for $B \geq 2$ its first coefficients include 1, 0 and a later 1. $\square$

Every pair $B \geq 2, t \geq 6\lfloor B/2 \rfloor$ supplies a counterexample to the original necessity assertion: the source condition would require $B \leq 1$. Thus the degree-nine example is not an isolated coincidence.

Corollary 6.4 (Minimality in the spacing-three branch). *Among counterexamples to the original necessity assertion with $r = 3$, at least six nontrivial ordinary factors are required, and the total polynomial degree is at least nine. Both bounds are attained by $(1+q)^6(1+q^3)$.*

Proof. A counterexample has no divisible factor and satisfies

$$Q + 1 < b \leq Q + 1 + 2\lfloor t/6 \rfloor.$$

Hence $t \geq 6$, so there are at least six factors of residue two, each nontrivial. Also $S = 3Q + t \geq 6$ and $b \geq 2$, whence $D = S + 3(b-1) \geq 9$. The displayed example realizes equality. $\square$

For $k \leq 5$, one necessarily has $t \leq 5$, so the correction $2\lfloor t/6 \rfloor$ vanishes. The corrected theorem therefore agrees exactly with the old spacing-three criterion throughout that range. This explains the relevance of the source's reported cutoff of five ordinary factors [1, p. 14]; it is not a numerical

Ordinary factors	Q	t	Old bound	Exact bound	Extra admissible b
$[2]_q^6$	0	6	1	3	2, 3
$[2]_q^{12}$	0	12	1	5	2, 3, 4, 5
$[5]_q^6$	6	6	7	9	8, 9
$[2]_q^5[4]_q$	1	5	2	2	None

Table 3: The exact bound means the largest b for which the product with $[b]_{q^3}$ is unimodal.

accident. The minimality assertion here is restricted to the $r = 3$ branch and is not a claim about every possible spacing.

The mechanism is arithmetic: modulo Φ_3 , a residue-two factor is $1 + q$, whose sixth power is one. Each complete group of six such factors allows two additional terms in the spaced factor. In contrast, for $r = 2$ the only nonzero residue is one, so no analogous correction exists.

7 An additional exact theorem: two shifted binomial rows

The smallest counterexample suggests isolating the family with $b = 2$. For this family there is a closed threshold for every spacing. The proof in this section is independent of the spacing-three residue table; it provides a further check on the special value six.

Theorem 7.1 (Sharp two-shift threshold). *For integers $r \geq 2$ and $n \geq 0$,*

$$(1 + q)^n(1 + q^r) \text{ is unimodal} \iff n \geq r^2 - 3. \quad (7.1)$$

Proof. Multiplication by $1 + q$ preserves SU by Lemma 2.1. It is consequently enough to prove unimodality at $n = r^2 - 3$ and failure at $n = r^2 - 4$. A unimodal polynomial at a smaller exponent would, on multiplication by a suitable power of $1 + q$, contradict the latter failure. For $r = 2$, the two boundary polynomials are respectively $1 + q + q^2 + q^3$ and $1 + q^2$, so the claim follows directly. Assume henceforth that $r \geq 3$.

For an exponent n , put $N = n + 1$ and define

$$H_n(j) = \binom{n}{j} - \binom{n}{j-1} = \frac{N-2j}{N} \binom{N}{j}. \quad (7.2)$$

Binomial coefficients outside their natural range are zero. Then $H_n(N-j) = -H_n(j)$. The coefficient difference at j in $(1 + q)^n(1 + q^r)$ is

$$H_n(j) + H_n(j-r). \quad (7.3)$$

Unimodality at $n = r^2 - 3$. Here $N = r^2 - 2$ and the polynomial degree $n + r$ is odd. For a first-half index $j \leq N/2$, both terms of (7.3) are nonnegative. For the remaining first-half indices, define

$$s = N + r - 2j, \quad u = j - r = \frac{N - r - s}{2}, \quad v = N - j = \frac{N - r + s}{2}.$$

Because $j > N/2$ and $j \leq (n + r - 1)/2$, the integer s is even and satisfies $2 \leq s < r$. Furthermore $0 \leq u < v < N/2$, so $H_n(u), H_n(v) > 0$. By reflection, (7.3) is $H_n(u) - H_n(v)$. It remains to prove

$$R_s := \frac{H_n(u)}{H_n(v)} = \frac{r+s}{r-s} \frac{\binom{N}{(N-r-s)/2}}{\binom{N}{(N-r+s)/2}} \geq 1. \quad (7.4)$$

At $s = 2$,

$$R_2 = \frac{(r+2)(N-r)(N-r+2)}{(r-2)(N+r)(N+r+2)} = 1,$$

because the numerator minus the denominator is

$$4N(N - r^2 + 2) = 0. \quad (7.5)$$

For an even s with $s + 2 < r$, the adjacent ratio is

$$\frac{R_{s+2}}{R_s} = \frac{(r + s + 2)(r - s)(N - r - s)(N - r + s + 2)}{(r + s)(r - s - 2)(N + r + s + 2)(N + r - s)}. \quad (7.6)$$

All factors in the denominator are positive. Expanding the numerator minus the denominator gives

$$4Nr(N - r^2 + s^2 + 2s + 2) = 4Nr s(s + 2) > 0, \quad (7.7)$$

where the equality uses $N = r^2 - 2$. Starting with $R_2 = 1$, induction over the even values of s proves (7.4). All first-half differences are nonnegative, and symmetry gives unimodality.

Failure at $n = r^2 - 4$. Now $N = r^2 - 3$ and the polynomial degree is even. At its central index

$$j = \frac{n + r}{2}, \quad u = j - r = \frac{N - r - 1}{2},$$

reflection turns (7.3) into $H_n(u) - H_n(u + 1)$. Using (7.2) and $\binom{N}{u+1} = \binom{N}{u}(N - u)/(u + 1)$, this difference is

$$\begin{aligned} H_n(u) - H_n(u + 1) &= \frac{\binom{N}{u}}{N} \left((r + 1) - (r - 1) \frac{N + r + 1}{N - r + 1} \right) \\ &= -\frac{4\binom{N}{u}}{N(N - r + 1)} < 0. \end{aligned} \quad (7.8)$$

The denominator is positive for $r \geq 3$. Thus unimodality fails at the previous exponent. The initial closure argument proves the full threshold (7.1). $\square$

For $r = 3$, Theorem 7.1 gives the threshold $n = 6$, in agreement with Corollary 6.3 for $B = 2$. Historical priority of Theorem 7.1 is not asserted here; its proof is included because it arose naturally from the counterexample and strengthens the resulting family analysis.

8 Exact computations and reproducibility

The proofs above establish the infinite statements. The accompanying `verify.py` separately checks finite cases using Python integers, without floating-point arithmetic or external packages. The run recorded in `verification_results.json` passed every check in Table 4.

8.1 What was checked

The exhaustive spacing-three test considers all multisets of ordinary parameters from $\{1, 2, 4, 5, 7, 8\}$, with zero through eight factors, and all b from one through the corrected bound plus two. The zero-factor cases check the harmless empty-product boundary. In this finite collection, 938 products violate the old necessity condition while satisfying the corrected theorem. Every predicted nonunimodal case is also checked at the explicit index (6.4); the drop is exactly -1 , not merely negative.

The separate random spacing-three experiment uses seed 1977, chooses k uniformly from $1, \dots, 16$, chooses each a_i from the integers $1, \dots, 69$ not divisible by three, and chooses b uniformly from one through the corrected bound plus eight. The seed is reset for this experiment so it is independent of how many preliminary tests are run.

Check	Number of cases
Independent schoolbook versus optimized multiplication	1,000
Centered peeling identity on an integer grid	3,003
Exhaustive spacing-three multiset products	30,539
Deterministic random spacing-three products, no divisible factor	30,000
Spacing-three products with a divisible factor	2,000
General-spacing sufficient condition	10,000
General-spacing necessary degree bound	10,000
Spacing-two necessary-and-sufficient criterion	5,000
Two-shift boundary polynomials, $2 \leq r \leq 60$	177
Exact two-shift negative central-difference formula	58
Spacing-three binomial-comb threshold samples	237

Table 4: Finite checks actually executed. Some families overlap; the counts are not numbers of distinct mathematical theorems or independent proofs.

The small counterexample is expanded by three different routes: sliding convolution, the binomial theorem, and direct enumeration of all $2^7 = 128$ choices of six weights one and one weight three. The generic unimodality checker tests the entire rise-then-fall pattern; it does not assume symmetry. A second checker uses symmetry and first-half differences.

A separate Wolfram Language evaluation, using the connected Wolfram kernel reporting version 15.0.1, confirmed the counterexample coefficients, the peeling identity, the residue table, the polynomial identities (7.5) and (7.7), and the two-shift boundary tests for $2 \leq r \leq 16$. The exact input and returned text are included as `verify.wl` and `wolfram_output.txt`. This is an independent computational check, not proof-assistant verification of the arguments.

8.2 Reproduction commands and output

Use Python 3.10 or later:

```
python3 verify.py --full --out verification_results.json
```

The command writes a JSON run summary, the small counterexample certificate, and a CSV table of spacing-three binomial thresholds. It returns a nonzero exit code if any check fails. For a shorter smoke test, omit `-full` and choose a different output filename to preserve the recorded full run.

Build this report with the supplied Makefile or run

```
latexmk -pdf -interaction=nonstopmode -halt-on-error report.tex
```

A TeX installation with the packages listed in the preamble is needed. No custom font files or proprietary packages are distributed.

8.3 Closed-form decision versus coefficient expansion

Theorem 6.1 gives a decision procedure requiring $O(k)$ integer-arithmetic operations: test divisibility, compute Q and t , and compare b with the bound. If the answer is negative, (6.4) gives the witness index directly. No expansion of the polynomial is needed to decide the question. The actual bit complexity depends on the sizes of the input integers.

The positive proof uses induction and the centered identity. Literally unrolling all Q peeling steps can be long when the parameters are encoded in binary; no claim of a uniformly short expanded certificate is being made. The full finite verifier expands coefficients because it is intended as an independent check, not as the fastest implementation of the classification theorem.

9 Scope and further targets

The exact counterexample disposes of the stated necessity assertion. The stronger results retain and prove its sufficient half, classify all spacing-two and spacing-three products, and identify a precise residue obstruction rather than a merely numerical discrepancy. The proofs explain both the first counterexample and the complete family that it begins.

The source's main q -Fibonomial question is different. This note neither proves nor disproves that main conjecture, and it does not challenge the source's established small-case theorems. For $r \geq 4$, a general classification of $(\prod_i [a_i]_q)[b]_{q^r}$ is not obtained here. In particular, the necessity question for two or three ordinary factors at those spacings is not settled by the present arguments.

A concrete continuation is to combine the centered reduction with the full vector of residue differences at larger r . Lemma 4.1 gives exact obstruction indices whenever the periodic tail enters the first half. Corollary 3.3 gives positive constructions from residue products. For $r = 3$, these two mechanisms meet exactly because $(1 + q)^6 \equiv 1 \pmod{\Phi_3(q)}$. For larger r , the residual factors $[s_i]_q$ range over more than 1 and $1 + q$, so the same one-parameter six-case reduction is no longer available. Determining when the positive and negative tests meet is a specific next task, not a conclusion asserted in this note.

A A short standalone certificate

The disproof itself does not depend on any of the stronger theorems. The following elementary Python fragment checks it using only the binomial theorem.

```
from math import comb

c = [
    (comb(6, j) if 0 <= j <= 6 else 0)
    + (comb(6, j-3) if 0 <= j-3 <= 6 else 0)
    for j in range(10)
]
assert c == [1, 6, 15, 21, 21, 21, 15, 6, 1]
assert c == c[::-1]
assert all(c[j-1] <= c[j] for j in range(1, 5))
a, b, r = [2]*6, 2, 3
assert not any(x % r == 0 for x in a)
assert b > 1 + sum(x/r for x in a)
assert len(a) <= 3 or r <= 3
```

The distributed full verifier uses explicit exceptions instead of `assert`, so running Python with optimization does not disable its checks. The fragment above is only a compact readable certificate.

References

- [1] Brendan B. Connelly, Ezekiel Ito, Thomas C. Martinez, Olha Shevchenko, and Kacey Yang. *Unimodality of q -Fibonomial coefficients for small cases*. arXiv:2605.12822v1 [math.CO], 2026. [Version-pinned arXiv record](#); [HTML](#); [PDF](#). Conjecture 5.4 and the adjacent testing statement, p. 14.